DIKRI ALI

+62 821 1178 4160 | dikri.ali_ti23@nusaputra.ac.id | LinkedIn | GitHub
Sukabumi, Indonesia

Software Engineering & Fullstack Developer specializing in AI & Computer Vision ecosystems. Passionate about bridging fullstack web engineering with intelligent systems to build end-to-end, scalable applications. Possesses hands-on experience in digitizing industrial processes during an IT engineering internship at PT Akebono Brake Astra Indonesia, alongside a strong track record of engineering RAG-based architectures (FastAPI, ChromaDB) and deploying YOLO computer vision models. Dedicated to delivering robust production systems, optimizing data ingestion pipelines, and creating interactive 3D digital environments.

PROFESIONAL EXPERIENCE

PT Akebono Brake Astra Indonesia – Jakarta, Indonesia **Feb 2026 - Sep 2026**

Fullstack Developer Intern

Key Responsibilities:

- Industrial System Architecture: Architected and engineered an end-to-end Dies Management System to digitize and automate dies-movement tracking across Astra Otoparts production lines.
- Computer Vision Implementation: Trained, evaluated, and fine-tuned YOLO-based object detection and segmentation models using Google Colab for manufacturing quality control.
- Dataset Engineering: Managed dataset acquisition, annotation, and preprocessing in Roboflow to improve the facility's automated visual inspection capabilities.
- 3D Asset & Digital Twin Modeling: Built detailed 3D digital twins of the manufacturing facility and interior industrial assets using Blender for interactive facility orientation and spatial planning.

Key Achievements:

- Process Digitization & Efficiency: Successfully replaced legacy manual, paper-based tracking workflows with a real-time digital management system across factory production lines, eliminating human recording errors.
- Smart Inspection Advancement: Accelerated shop-floor automated visual inspection development by training high-precision object detection models on custom production datasets.
- Factory Operational Visibility: Enhanced facility orientation and spatial planning efficiency by delivering an interactive, high-fidelity 3D digital twin of the factory layout and interior assets.
- End-to-End Delivery: Successfully delivered production-ready software and dataset pipelines on schedule while collaborating closely with the internal IT department.

PROJECT EXPERIENCE

ChatbotSunda – RAG-Based Sundanese Language & Culture Chatbot

Personal Project | *FastAPI, Vue 3, ChromaDB, Redis, Docker*

- Full-Stack RAG Engineering: Designed and built an end-to-end Retrieval-Augmented Generation (RAG) platform featuring a Vue 3 frontend, a FastAPI backend with Server-Sent Events (SSE) streaming, and an internal operational dashboard for ingestion and retrieval debugging.

- Hybrid Retrieval Pipeline: Engineered a hybrid search pipeline integrating multilingual semantic search (ChromaDB) and BM25 keyword search, augmented with reranking and Maximal Marginal Relevance (MMR) diversity filtering.
- Data & Ingestion Pipeline: Implemented an automated document ingestion pipeline with hash-based deduplication, content categorization, and dynamic chunking.
- Production Reliability: Integrated Redis-backed semantic caching, rate limiting, and session history management for enterprise-grade API performance.

Interactive 3D Portfolio – BMS/HUD-Style Personal Website

Personal Project | *Blender, Blender MCP, Hyper3D Rodin*

- 3D Web Experience: Designed and developed an interactive 3D portfolio centered on a custom Blender-modeled architecture styled with a Building Management System (BMS)/HUD aesthetic.
- Interactive Navigation: Implemented dynamic camera flight paths and interactive 3D scene triggers linked to UI navigation.
- AI-Assisted Asset Pipeline: Integrated an AI-assisted 3D generation pipeline (Blender MCP + Hyper3D Rodin) to accelerate 3D spatial asset creation and scene iteration.

EDUCATION

Universitas Nusa Putra – Sukabumi, Indonesia

Bachelor of Informatics Engineering (Software Engineering) | GPA: 3.68 / 4.00 | Seventh-semester Student

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SKILLS

Hard Skills

- Programming Languages: Python, PHP, JavaScript
- Frameworks & Libraries: Laravel, FastAPI, React, Vue.js
- Databases & Caching: PostgreSQL, MySQL, ChromaDB, Redis
- AI & Computer Vision: YOLO, OpenCV, OCR, Roboflow, RAG Frameworks
- Development & DevOps: REST API, Git, Docker
- 3D Design & Tools: Blender, Blender MCP, Hyper3D Rodin

Soft Skills: Problem Solving, Analytical Thinking, Technical Communication, Adaptability, System Architecture Design

Interest: Software Engineering, Fullstack Development, AI & Computer Vision, RAG Systems, 3D Web & Digital Twins